

SIMATS ENGINEERING

Assessment Tool 1 — Annexure A: Design Task

Course Name	Artificial Intelligence	Course Code	CSA17
Duration	1 Hour	Total Marks	50
Questions	5	Weightage	50%

Course Outcome (CO5): Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. *(BTL 5 — Evaluate/Create)*

PROBLEM STATEMENTS

The following five design problems require the student to architect a scalable, distributed Artificial Intelligence system. Each solution must explicitly use **Apache PySpark** and **RDD (Resilient Distributed Dataset) operations** for large-scale data processing, and must incorporate an **intelligent agent architecture** that perceives data, reasons over it, and acts (adaptively, in real time) to fulfil the system's objective.

Question 1: Hybrid AI Recommendation Engine Design

Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

Question 2: Smart Disaster Detection System

Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

Question 3: AI-Based Fraud Detection Framework

Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

Question 4: Autonomous Supply Chain Optimization System

Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support large-scale data processing

and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

Question 5: Personalized Learning Analytics Platform

Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

EVALUATION RUBRIC (SUMMARY)

Criteria	Weight
Problem Understanding & Requirement Analysis	10%
System Architecture Design	20%
Use of PySpark & RDD Operations	15%
Intelligent Agent Integration	15%
Scalability & Distributed Computing Design	10%
Data Flow & Processing Pipeline	10%
Innovation & Creativity	10%
Clarity, Presentation & Documentation	10%

Note: Full descriptive rubric levels (Excellent / Good / Average / Poor) for each criterion are provided in the accompanying Report document.